PROJECT REPORT

Computer Organization and Assembly Language Lab (CEL-324)

PROJECT TITLE: <u>CAFÉ MANAGEMENT SYSTEM</u>

BS(CS)-3A

Group Members

Name	Enrollment
1. MAHA SHAHID	02-134222-103
2. NAHIN FATIMA	02-134222-102
3. ALISHBA SIRAJ	02-134222-005

Submitted to:

Ma'am Mehwish Saleem

BAHRIA UNIVERSITY KARACHI CAMPUS

Department of Computer Science

ABSTRACT

The Cafe Management System in assembly language is a comprehensive program designed to enhance the ordering process within a cafe environment. This system offers users an intuitive interface to select items from hi tea, brunch, and buffet menus, with each menu item associated with a predefined price. The program maintains a dynamic running total of the selected items and their corresponding prices, optimizing the order entry experience. Robust error-checking mechanisms are integrated to ensure the system handles invalid inputs effectively, providing users with feedback for smooth operation. Upon completion of the order, the system calculates the total bill by summing up the prices of the selected items, facilitating accurate and efficient billing. This assembly language implementation prioritizes efficiency in processing and memory usage, contributing to an optimized cafe management solution that streamlines customer interactions and improves overall operational effectiveness.

INTRODUCTION:

The Cafe Management System implemented in assembly language represents a sophisticated solution tailored to streamline the order processing and billing procedures within a cafe setting. With a focus on user-friendly interactions, this system enables customers to effortlessly select items from hi tea, brunch, and buffet menus. Each menu item is intricately associated with a predefined price, and the program dynamically calculates a running total as users build their orders. The implementation prioritizes efficiency by leveraging assembly language, ensuring optimized processing and memory usage.

Beyond its user interface, the system incorporates robust error-checking mechanisms to handle potential input discrepancies, enhancing the reliability of the order entry process. Once an order is complete, the system seamlessly calculates the total bill, providing an accurate and efficient billing experience for both customers and cafe staff.

This Cafe Management System stands out not only for its functional prowess but also for its commitment to enhancing overall operational effectiveness within a cafe environment. Its assembly language foundation ensures a balance between performance and resource utilization, making it a valuable tool for cafes aiming to improve customer service and streamline their daily operations.

PROBLEM STATEMENT:

Cafes often face challenges in efficiently managing customer orders and billing processes, leading to potential errors, delays, and inefficiencies in their daily operations. The absence of a dedicated and optimized system for order processing, especially one tailored to handle diverse menus such as hi tea, brunch, and buffet, contributes to these challenges. Manual order calculations and the lack of robust error-checking mechanisms further exacerbate the risk of inaccuracies, impacting customer satisfaction and hindering overall operational effectiveness.

To address these issues, there is a pressing need for a Cafe Management System implemented in assembly language. This system should provide an intuitive user interface for customers to seamlessly select items from different menus, accurately calculate the total bill based on item prices, and incorporate error-handling mechanisms to enhance reliability. The proposed solution aims to optimize resource utilization through its assembly language implementation, ensuring efficient processing and memory usage. This system will not only streamline the order entry and billing processes but also contribute to an improved customer experience and increased operational efficiency within cafes.

METHODOLOGY:

The methodology for developing the Cafe Management System in assembly language involves a step-by-step process. It begins with requirements analysis, identifying user interface design, menu categorization, and error-handling needs. The assembly language is selected based on compatibility and optimization for processing and memory. The user interface is then designed to allow easy navigation through menus, with careful consideration for item selection and feedback. Implementation includes creating a data structure for menu items and prices, along with functions for calculations. Robust error-checking is integrated, and total bill calculation is optimized for accuracy and efficiency.

Thorough testing, debugging, and performance evaluations ensure the system's reliability. User acceptance testing gathers feedback from potential users, guiding necessary adjustments. Comprehensive documentation, including manuals and technical guides, is prepared to facilitate future maintenance and updates. This methodology ensures the development of an efficient and user-friendly Cafe Management System tailored to cafe operational needs.

PROJECT SCOPE:

The scope of the Cafe Management System project in assembly language encompasses the design, development, implementation, and documentation of a comprehensive system to streamline order processing and billing within a cafe setting. The project will include the following key components:

- ✓ **User Interface:**

Design an intuitive and user-friendly interface allowing customers to select items from hi tea, brunch, and buffet menus.

Implement input mechanisms for item selection and ensure clear and informative feedback to users.

- ✓ **Menu Management:**

Categorize menu items into hi tea, brunch, and buffet, associating each item with a predefined price.

Create a data structure to efficiently manage menu items and their corresponding prices.

- ✓ **Order Processing:**

Develop functions or procedures in assembly language to handle menu item selection, calculate the running total, and validate user inputs.

Integrate error-checking mechanisms to gracefully handle invalid selections and provide informative error messages.

- ✓ **Billing System:**

Implement functions for accurate total bill calculation based on the selected items and their associated prices.

The project scope aims to deliver a robust and efficient Cafe Management System implemented in assembly language, addressing the specific needs of cafes by optimizing order processing, enhancing user experience, and ensuring accurate billing.

CODE:

```
.model large
print macro m
   mov ah, 09
   lea dx, m
   int 21h
endm

.stack 100h
.data
M1 DB 10,13,10,13,'                    Welcome to Cafe Lola =D                    $'
ST DB 10,13,10,13,'       Cafe owned by three girls, different personalities under one roof $'
M2 DB 10,13,10,13,'Would you like to have:  $'

M3 DB 10,13,'        1.Brunch       $'
M4 DB 10,13,'        2.Hi-Tea        $'
M5 DB 10,13,'        3.Buffet       $'


str db 10,13,'3.Continue with the same menu.      $'

M8 DB 10,13,10,13,'Enter the number associated with the dish you would like to have: $'
M6 DB 10,13,10,13,'Enter the number associated with the age group you prefer to get buffet for: $'

;BRUNCH
M9 DB 10,13,'      1.Creme Brulee French Toast       Rs 10            $'
```

M10 DB 10,13,' 2.Choco Cinnamon Rolls (3 pieces) Rs 10 $'

M11 DB 10,13,' 3.Quiche Rs 10 $'

M12 DB 10,13,' 4.Acai Bowl Rs 10 $'

M13 DB 10,13,' 5.Granola Bowl with choice of fruits Rs 20 $'

M14 DB 10,13,' 6.Crossiant Sandwiches Rs 20 $'

M15 DB 10,13,' 7.Assorted Bread Basket with drink Rs 10 $'

M16 DB 10,13,' 8.Seasonal Fruit Smoothies Rs 20 $'

M17 DB 10,13,' 9.Coffee Rs 60 $'

;Hi-Tea

M25 DB 10,13,' 1.Dynamit Shrimps (6 pieces) Rs 90 $'

M26 DB 10,13,' 2.Korean Fried Chicken (4 pieces) Rs 90 $'

M27 DB 10,13,' 3.Dumplings (6 pieces) Rs 30 $'

M28 DB 10,13,' 4.Beijing Chicken Drummets (4 pieces) Rs 90 $'

M29 DB 10,13,' 5.Choco loco Mousse Rs 90 $'

M30 DB 10,13,' 6.Tea Rs 10 $'

M31 DB 10,13,' 7.Pudding with Fresh Fruits Rs 30 $'

M32 DB 10,13,' 8.Coffee Rs 30 $'

M33 DB 10,13,' 9.Chiller Rs 30 $'

;Buffet

M34 DB 10,13,' 1.Kids (under 10years old) Rs 30 $'

M35 DB 10,13,' 2.Students (above 10 but under 20) Rs 60 $'

M36 DB 10,13,' 3.Adults Rs 90 $'

M37 DB 10,13,' 4.Senior Citizens Rs 80 $'

;INVALID

```
M55 DB 10,13,10,13,'  INVALID option selected $'

M56 DB 10,13,'      Try Again $'


M57 DB 10,13,10,13,'Enter the number associated with the dish you would like to have: $'

M58 DB 10,13,'Quantity: $'

M59 DB 10,13,'Total Bill:RS $'


QUANTITY DB ?

price db ?


M60 DB 10,13,10,13,'1.Go Back to Main Menu$'

M61 DB 10,13,'2.EXIT$'


M62 DB 10,13,'   Due to a lot of customers at the moment sorry u can only order two things for now. $'


SEJ DB 10,13,10,13,' $'


.code

MAIN PROC

   MOV AX,@DATA

   MOV DS,AX

  TOP:


   print M1

   print ST

   print SEJ ;NEWLINE


   mov price,'0'
```

```
print M3   ; brunch
print M4
print M5

print M2 ; choice



MOV AH,1    ; input
INT 21H
MOV BH,AL
SUB BH,48

CMP BH,1
JE BRUNCH

CMP BH,2
JE HiTea

CMP BH,3
JE BUFFET
JMP INVALID

BRUNCH:
        ;BRUNCH STARTS

LEA DX,SEJ ;NEWLINE
MOV AH,9
INT 21H
```

```
print M9
print M10
print M11
print M12
print M13
print M14
print M15
print M16
print M17
print M57

MOV AH,1
INT 21H
MOV BL,AL
SUB BL,48

CMP BL,1
JE TEN

CMP BL,2
JE TEN

CMP BL,3
JE TEN

CMP BL,4
JE TEN
```

```
CMP BL,5
JE TWENTY

CMP BL,6
JE TWENTY

CMP BL,7
JE TEN

CMP BL,8
JE TWENTY

CMP BL,9
JE SIXTY

JMP INVALID

FOURTY:
MOV BL,4
LEA DX,M58
MOV AH,9
INT 21H

MOV AH,1
INT 21H
SUB AL,48
```

```
MUL BL
AAM

MOV CX,AX
ADD CH,48
ADD CL,48


LEA DX,M59
MOV AH,9
INT 21H

MOV AH,2
MOV DL,CH
INT 21H


MOV DL,CL
INT 21H

MOV DL,'0'
INT 21H

;FOR /- PRINT
MOV DL,47
INT 21H
MOV DL,45
INT 21H
```

```
;GO BACK TO MAIN MENU

LEA DX,M60
MOV AH,9
INT 21H

LEA DX,M61
MOV AH,9
INT 21H

lea Dx,str
mov ah,09    ;for going same menu
int 21h



LEA DX,M2
MOV AH,9
INT 21H          ;MAIN MENU

MOV AH,1
INT 21H
SUB AL,48




CMP AL,1
JE TOP
cmp Al,2
JE BRUNCH
```

```
JMP EXIT

FIFTY:
MOV BL,4
LEA DX,M58
MOV AH,9
INT 21H

MOV AH,1
INT 21H
SUB AL,48

MUL BL
AAM

MOV CX,AX
ADD CH,48
ADD CL,48



LEA DX,M59
MOV AH,9
INT 21H

MOV AH,2
MOV DL,CH
INT 21H
```

```
MOV DL,CL
INT 21H

MOV DL,'0'
INT 21H



;FOR /- PRINT
MOV DL,47
INT 21H
MOV DL,45
INT 21H



;GO BACK TO MAIN MENU

LEA DX,M60
MOV AH,9
INT 21H

LEA DX,M61
MOV AH,9
INT 21H

LEA DX,M2
MOV AH,9
INT 21H

MOV AH,1
```

```
INT 21H
SUB AL,48    ;MAIN  MENU

CMP AL,1
JE TOP

JMP EXIT

SEVENTY:
MOV BL,7
LEA DX,M58
MOV AH,9
INT 21H



MOV AH,1
INT 21H
SUB AL,48



MUL BL
AAM

MOV CX,AX
ADD CH,48
ADD CL,48
```

```
LEA DX,M59
MOV AH,9
INT 21H

MOV AH,2
MOV DL,CH
INT 21H



MOV DL,CL
INT 21H

MOV DL,'0'
INT 21H



;FOR /- PRINT
MOV DL,47
INT 21H
MOV DL,45
INT 21H

;GO BACK TO MAIN MENU

LEA DX,M60
MOV AH,9
INT 21H
```

```
LEA DX,M61
MOV AH,9
INT 21H

LEA DX,M2
MOV AH,9
INT 21H      ;MAIN MENU

MOV AH,1
INT 21H
SUB AL,48



CMP AL,1
JE TOP

JMP EXIT

EIGHTY:
MOV BL,8
LEA DX,M58
MOV AH,9
INT 21H

MOV AH,1
INT 21H
SUB AL,48

MUL BL
```

```
AAM

MOV CX,AX
ADD CH,48
ADD CL,48

LEA DX,M59
MOV AH,9
INT 21H

MOV AH,2
MOV DL,CH
INT 21H

MOV DL,CL
INT 21H

MOV DL,'0'
INT 21H

;FOR /- PRINT
MOV DL,47
INT 21H
MOV DL,45
INT 21H

;GO BACK TO MAIN MENU

LEA DX,M60
```

```
MOV AH,9
INT 21H

LEA DX,M61
MOV AH,9
INT 21H

LEA DX,M2    ;MAIN MENU
MOV AH,9
INT 21H

MOV AH,1
INT 21H
SUB AL,48

CMP AL,1
JE TOP
JMP EXIT
JMP EXIT

 HiTea:
               ;Hi-Tea

LEA DX,SEJ ;NEWLINE
MOV AH,9
INT 21H

print M25
print M26
```

```
print M27
print M28
print M29
print M30
print M31
print M32
print M33
print M57


MOV AH,1
INT 21H
MOV BL,AL
SUB BL,48

CMP BL,1
JE NINETY

CMP BL,2
JE NINETY

CMP BL,3
JE THIRTY

CMP BL,4
JE NINETY

CMP BL,5
JE NINETY
```

```
CMP BL,6
JE TEN

CMP BL,7
JE THIRTY

CMP BL,8
JE THIRTY

CMP BL,9
JE THIRTY

JMP INVALID

TEN:
MOV BL,1
LEA DX,M58
MOV AH,9
INT 21H

MOV AH,1
INT 21H
SUB AL,48

MUL BL
AAM

MOV CX,AX
```

```
ADD CH,48
ADD CL,48

print M59

MOV AH,2
MOV DL,CH
INT 21H

MOV DL,CL
INT 21H

MOV DL,'0'
INT 21H

;FOR /- PRINT

MOV DL,47
INT 21H
MOV DL,45
INT 21H

;GO BACK TO MAIN MENU

LEA DX,M60
MOV AH,9
INT 21H
```

```
LEA DX,M61
MOV AH,9
INT 21H

lea dx,str
mov ah,09
int 21h

LEA DX,M2
MOV AH,9
INT 21H


MOV AH,1
INT 21H
SUB AL,48

CMP AL,1
JE TOP
cmp Al,3
JE BRUNCH
JMP EXIT

TWENTY:
MOV BL,2
LEA DX,M58
MOV AH,9
INT 21H
```

```
MOV AH,1
INT 21H
SUB AL,48

MUL BL
AAM

MOV CX,AX
ADD CH,48
ADD CL,48


LEA DX,M59
MOV AH,9
INT 21H

MOV AH,2
MOV DL,CH
INT 21H

MOV DL,CL
INT 21H

MOV DL,'0'
INT 21H

;FOR /- PRINT
MOV DL,47
INT 21H
```

```
MOV DL,45
INT 21H

;GO BACK TO MAIN MENU
LEA DX,M60
MOV AH,9
INT 21H

LEA DX,M61
MOV AH,9
INT 21H

LEA DX,M2
MOV AH,9
INT 21H

MOV AH,1
INT 21H
SUB AL,48


CMP AL,1
JE TOP
JMP EXIT

THIRTY:
MOV BL,3
LEA DX,M58
MOV AH,9
```

```
INT 21H

MOV AH,1
INT 21H
SUB AL,48

MUL BL
AAM

MOV CX,AX
ADD CH,48
ADD CL,48

LEA DX,M59
MOV AH,9
INT 21H

MOV AH,2
MOV DL,CH
INT 21H



MOV DL,CL
INT 21H

MOV DL,'0'
INT 21H

;FOR /- PRINT
```

```
MOV DL,47
INT 21H
MOV DL,45
INT 21H

;GO BACK TO MAIN MENU

LEA DX,M60
MOV AH,9
INT 21H

LEA DX,M61
MOV AH,9
INT 21H

lea dx,str
mov ah,09
int 21h

LEA DX,M2
MOV AH,9
INT 21H

MOV AH,1
INT 21H
SUB AL,48

CMP AL,1
JE TOP
```

```
cmp al,3
je HiTea
JMP EXIT

SIXTY:

MOV BL,6

LEA DX,M58
MOV AH,9
INT 21H

MOV AH,1
INT 21H
SUB AL,48

MUL BL
AAM

MOV CX,AX
ADD CH,48
ADD CL,48

LEA DX,M59
MOV AH,9
INT 21H

MOV AH,2
MOV DL,CH
```

```
INT 21H



MOV DL,CL
INT 21H

MOV DL,'0'
INT 21H

;FOR /- PRINT
MOV DL,47
INT 21H
MOV DL,45
INT 21H

;GO BACK TO MAIN MENU

LEA DX,M60
MOV AH,9
INT 21H

LEA DX,M61
MOV AH,9
INT 21H           ;MAIN MENU

LEA DX,M2
MOV AH,9
INT 21H
```

```
lea dx,str
mov ah,09
int 21h

MOV AH,1
INT 21H
SUB AL,48

CMP AL,1
JE TOP
cmp al,3
je HiTea
JMP EXIT

NINETY:
MOV BL,9

LEA DX,M58
MOV AH,9
INT 21H

MOV AH,1
INT 21H
SUB AL,48

MUL BL
AAM

MOV CX,AX
```

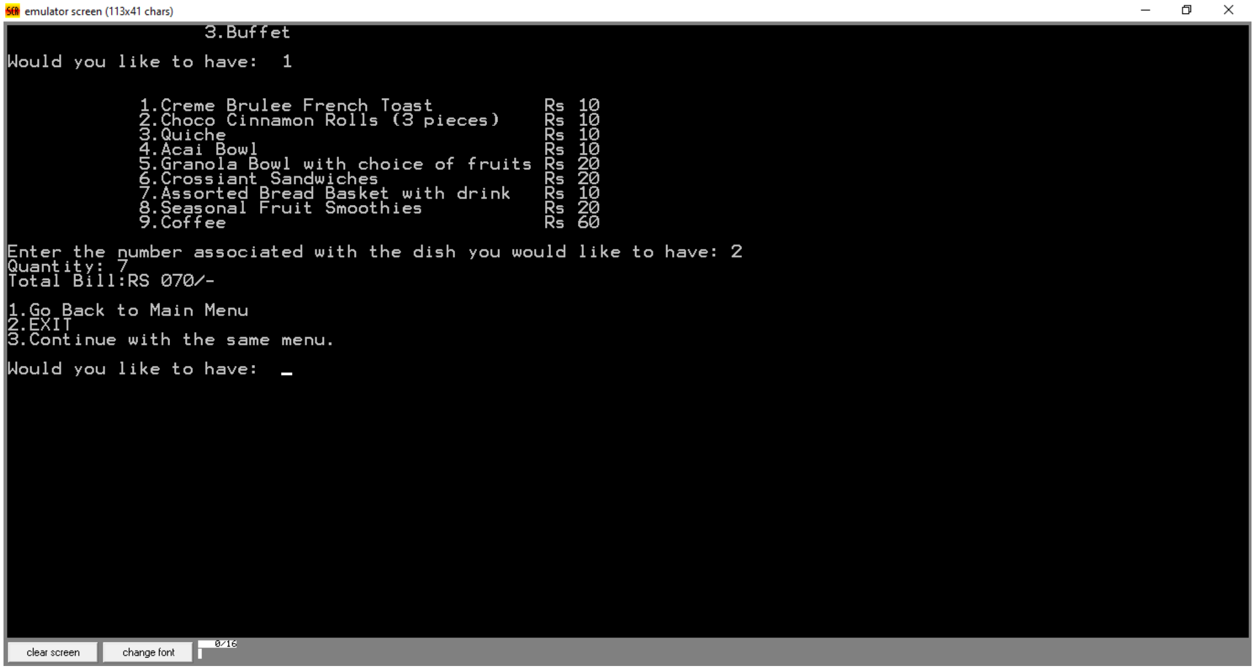
```
ADD CH,48
ADD CL,48



LEA DX,M59
MOV AH,9
INT 21H

MOV AH,2
MOV DL,CH
INT 21H
MOV DL,CL
INT 21H

MOV DL,'0'
INT 21H

;FOR /- PRINT
MOV DL,47
INT 21H
MOV DL,45
INT 21H

;GO BACK TO MAIN MENU

LEA DX,M60
MOV AH,9
INT 21H
```

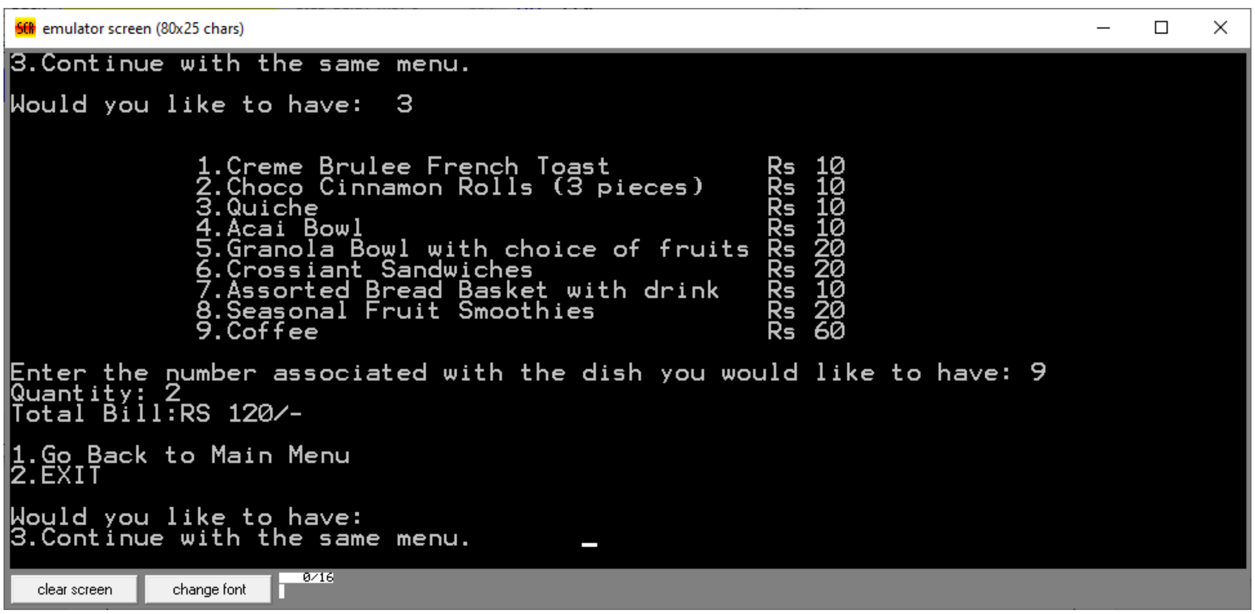
```
LEA DX,M61
MOV AH,9
INT 21H

LEA DX,M2
MOV AH,9
INT 21H

MOV AH,1
INT 21H
SUB AL,48

CMP AL,1
JE TOP

JMP EXIT
 JMP EXIT

 BUFFET:
        ;BUFFET STARTS

LEA DX,SEJ ;NEWLINE
MOV AH,9
INT 21H

print M34
print M35
print M36
print M37
```

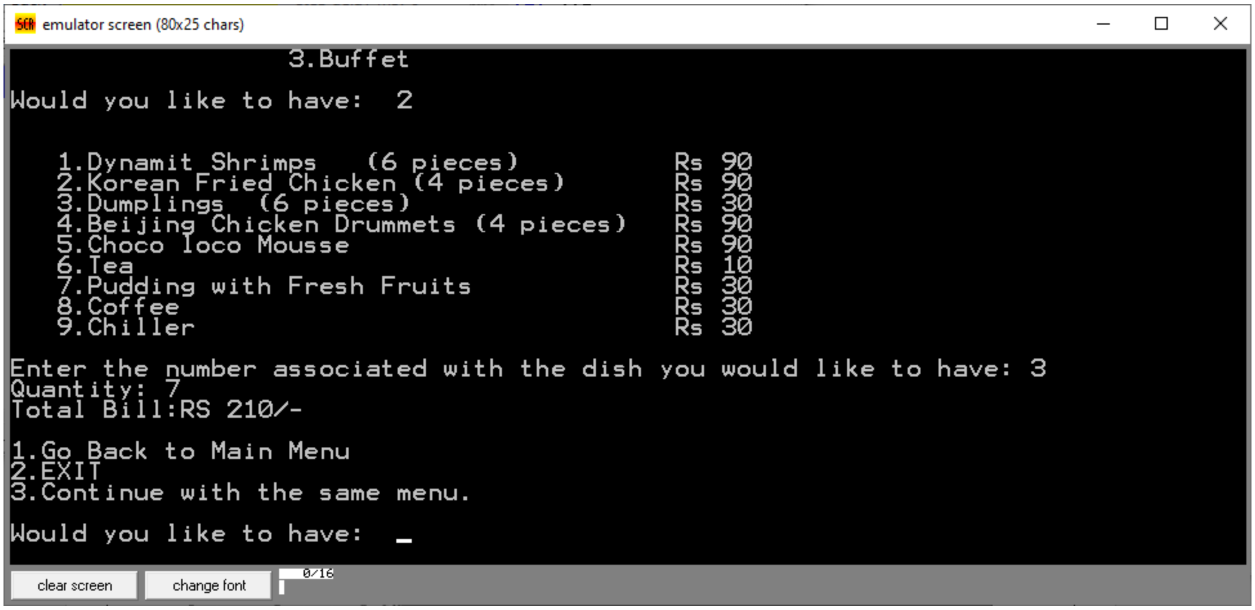
```
print M6

MOV AH,1
INT 21H
MOV BL,AL
SUB BL,48

CMP BL,1
JE THIRTY

CMP BL,2
JE SIXTY

CMP BL,3
JE NINETY
CMP BL,4
JE EIGHTY
JMP INVALID


INVALID:
 LEA DX,M55
 MOV AH,9
 INT 21H
 LEA DX,M56
 MOV AH,9
 INT 21H
 JMP EXIT
  EXIT:
```

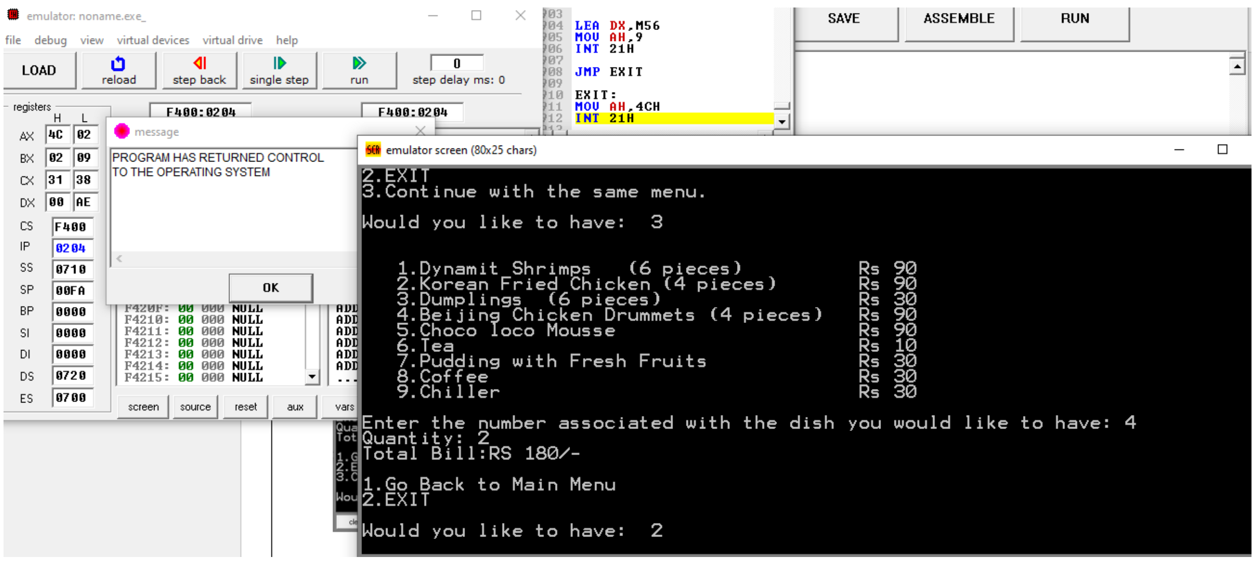
```
MOV AH,4CH
INT 21H
end main
```

OUTPUT:


emulator screen (113x41 chars)
3.Buffet
Would you like to have: 1
1.Creme Brulee French Toast Rs 10
2.Choco Cinnamon Rolls (3 pieces) Rs 10
3.Quiche Rs 10
4.Acai Bowl Rs 10
5.Granola Bowl with choice of fruits Rs 20
6.Crossiant Sandwiches Rs 20
7.Assorted Bread Basket with drink Rs 10
8.Seasonal Fruit Smoothies Rs 20
9.Coffee Rs 60
Enter the number associated with the dish you would like to have: 2
Quantity: 7
Total Bill:RS 070/-
1.Go Back to Main Menu
2.EXIT
3.Continue with the same menu.
Would you like to have:
clear screen
change font
0/16


emulator screen (80x25 chars)
3.Continue with the same menu.
Would you like to have: 3
1.Creme Brulee French Toast Rs 10
2.Choco Cinnamon Rolls (3 pieces) Rs 10
3.Quiche Rs 10
4.Acai Bowl Rs 10
5.Granola Bowl with choice of fruits Rs 20
6.Crossiant Sandwiches Rs 20
7.Assorted Bread Basket with drink Rs 10
8.Seasonal Fruit Smoothies Rs 20
9.Coffee Rs 60
Enter the number associated with the dish you would like to have: 9
Quantity: 2
Total Bill:RS 120/-
1.Go Back to Main Menu
2.EXIT
Would you like to have:
3.Continue with the same menu.
clear screen
change font
0/16





FUTURE DEVELOPMENT:

The future development of the cafe management system, which currently handles customer orders and bill generation, could include key enhancements. Integrating a customer relationship management (CRM) system would personalize the experience through detailed customer profiles and loyalty programs. Advanced inventory management features would optimize stock levels, and mobile application compatibility would cater to modern ordering preferences. Analytical tools for sales trends and reporting features would offer valuable insights, aiding strategic decision-making. Payment integration with diverse methods and potential POS system integration would streamline transactions, enhancing accuracy and financial management. These developments aim to position the cafe management system as a comprehensive, customer-focused, and technologically advanced solution for optimized operational efficiency and customer satisfaction.

CONCLUSION

In summary, the envisioned future developments for the cafe management system aim to enhance customer experience and operational efficiency. Integrating CRM functionalities will personalize interactions through loyalty programs and detailed customer profiles. Advanced inventory management and mobile application compatibility address modern demands for efficient stock monitoring and convenient, contactless ordering. Analytical tools and reporting features offer valuable insights for strategic decision-making, optimizing menus and operations. Additionally, proposed payment integration and potential collaboration with Point of Sale (POS) systems streamline transactions, improving accuracy and financial management. These advancements position the cafe management system as a sophisticated, customer-centric, and technologically advanced solution to meet the evolving needs of both cafe owners and patrons.